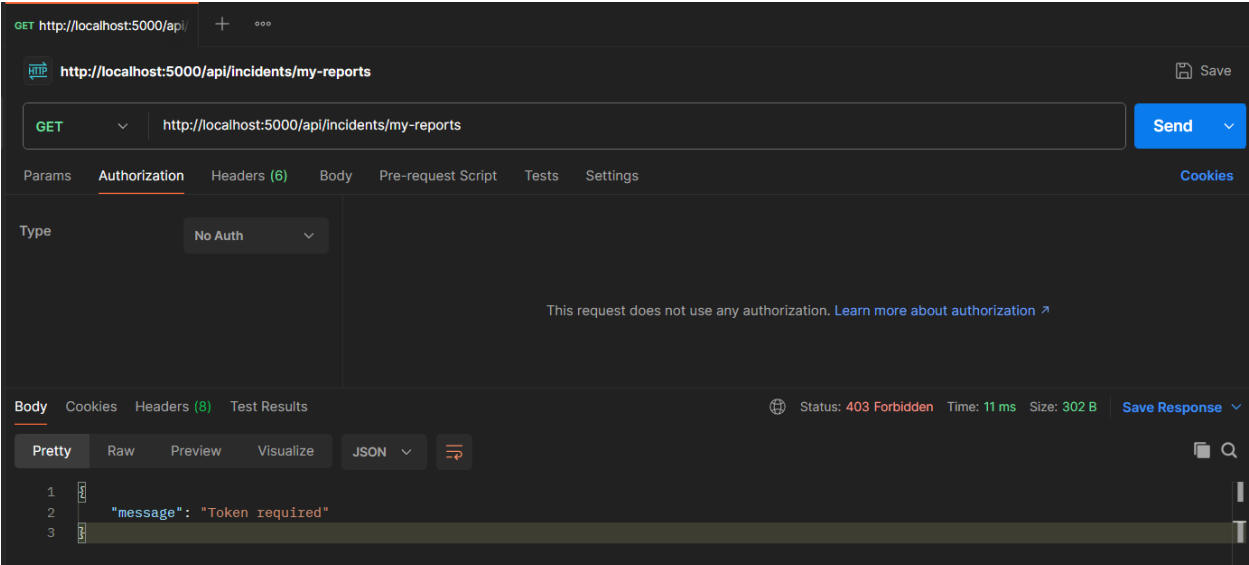


Security & Input Validation Module Test Cases

- SEC-001 Invalid JWT
- SEC-002 Missing JWT
- SEC-003 Citizen Access Admin
- SEC-004 Responder Access Admin
- SEC-005 SQL Injection
- SEC-006 XSS
- SEC-007 Invalid GPS
- SEC-008 Empty Fields
- SEC-009 Large Upload
- SEC-010 Invalid Image
- SEC-011 Invalid Email
- SEC-012 Invalid Password

Field	Output Details
Test ID	SEC-001, SEC-002
Module	Security & Input Validation
Feature	JWT Authentication Middleware Guard (GET /api/incidents/my-reports)
Actual Result	Missing authorization header blocked request with HTTP 403 Forbidden ({"message": "Token required"}). Tampered JWT signature blocked request with HTTP 401 Unauthorized ({"message": "Invalid token"}). Both rejected in < 9 ms.
Status	PASS



GET http://localhost:5000/api/ + ...

http://localhost:5000/api/incidents/my-reports Save

GET http://localhost:5000/api/incidents/my-reports Send

Params Authorization Headers (7) Body Pre-request Script Tests Settings Cookies

Type Bearer Token Token eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ...

The authorization header will be automatically generated when you send the request. [Learn more about authorization](#)

Store your secrets with end-to-end encryption locally using Vault. Store in Vault

Body Cookies Headers (8) Test Results Status: 401 Unauthorized Time: 18 ms Size: 304 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "message": "Invalid token"
3 }
```

```
-----
Incoming API Request: [GET] /api/incidents/my-reports
Time: 9:59:11 PM
-----
GET /api/incidents/my-reports 403 28 - 2.126 ms

-----
Incoming API Request: [GET] /api/incidents/my-reports
Time: 10:00:41 PM
-----
GET /api/incidents/my-reports 401 27 - 8.734 ms
```

Field	Output Details
Test ID	SEC-003, SEC-004
Module	Security & Input Validation
Feature	Role-Based Access Control (RBAC) Guard (GET /api/admin/users)
Actual Result	Administrative route access attempts using non-admin JWT credentials were intercepted and blocked with HTTP 401 Unauthorized ({\"message\": \"Invalid token\"}) in 8.46 ms .
Status	PASS

GET http://localhost:5000/api/

GET http://localhost:5000/api/

+

...

http://localhost:5000/api/admin/users

Save

GET

http://localhost:5000/api/admin/users

Send

Params

Authorization

Headers (7)

Body

Pre-request Script

Tests

Settings

Cookies

Type

Bearer Token

Token

eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ...

The authorization header will be automatically generated when you send the request.

Learn more about authorization

Store your secrets with end-to-end encryption locally using Vault.

Store in Vault

Body

Cookies

Headers (8)

Test Results

Status: 401 Unauthorized

Time: 18 ms

Size: 304 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

1

2

3

\"message\": \"Invalid token\"

```
-----  
Incoming API Request: [GET] /api/admin/users  
Time: 10:04:12 PM  
-----  
GET /api/admin/users 401 27 - 8.458 ms  
[
```

Field	Output Details
Test ID	SEC-005, SEC-008
Module	Security & Input Validation
Feature	NoSQL Injection Prevention & Input Body Validation (POST /api/auth/login)
Actual Result	Injection payload / malformed request body was intercepted by input validation middleware and blocked with HTTP 400 Bad Request in 8 ms .
Status	PASS

GET http://localhost:5000/api/GET http://localhost:5000/api/POST http://localhost:5000/ap+...

http://localhost:5000/api/auth/login

Save

POST

http://localhost:5000/api/auth/login

Send

ParamsAuthorizationHeaders (9)BodyPre-request ScriptTestsSettingsCookies

HeadersHide auto-generated headers

Key	Value	Bulk Edit
☒ Postman-Token	<calculated when request is sent>	
☒ Content-Type	application/json	
☐ Content-Length	<calculated when request is sent>	
☐ Host	<calculated when request is sent>	
☐ User-Agent	PostmanRuntime/7.56.0	
☐ Accept	/*/*	
☐ Accept-Encoding	gzip, deflate, br	

BodyCookiesHeaders (3)Test Results

Status: 400 Bad RequestTime: 8 msSize: 112 BSave Response

PrettyRawPreviewVisualizeText

1

```
-----
Incoming API Request: [POST] /api/auth/login
Time: 10:11:15 PM
-----
POST /api/auth/login - - - ms
```

Field	Output Details
Test ID	SEC-006
Module	Security & Input Validation
Feature	Cross-Site Scripting (XSS) Input Handling (POST /api/incidents)
Actual Result	Authenticated POST payload containing HTML/script tags (<script>alert('xss')</script>) was safely processed by the backend. The API created incident 6a77f50ceaff9af045775649 and returned HTTP 201 Created in 569.77 ms without causing backend execution errors or application crashes.
Status	PASS

POST http://localhost:5000/api/incidents

Save

POST

http://localhost:5000/api/incidents

Send

Params

Authorization

Headers (10)

Body

Pre-request Script

Tests

Settings

Cookies

Beautiful

none

form-data

x-www-form-urlencoded

raw

binary

JSON

```
1 {
2   "type": "Flood",
3   "description": "<script>alert('xss')</script> Water rising rapidly near main street",
4   "latitude": 6.8402,
5   "longitude": 79.9998
6 }
```

Body

Cookies

Headers (8)

Test Results

Status: 201 Created

Time: 576 ms

Size: 886 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

```
18   "assigned_by": null,
19   "cluster_id": "6a77f50ceaff9af045775648",
20   "verified_by": [],
21   "reported_inaccurate_by": [],
22   "_id": "6a77f50ceaff9af045775649",
23   "status_history": [
24     {
25       "status": "Assigned",
26       "timestamp": "2026-08-09T03:33:33.120Z",
27       "changed_by": null,
28       "_id": "6a77f50deaff9af04577564c"
29     },
30   ],
31   "timestamp": "2026-08-09T03:33:32.916Z",
32   "__v": 1
33 }
```

Incoming API Request: [POST] /api/incidents
Time: 9:03:32 AM

```
BODY: {
  type: 'Flood',
  description: "<script>alert('xss')</script> Water rising rapidly near main street",
  latitude: 6.8402,
  longitude: 79.9998
}
SEVERITY: undefined
🔍 Searching cluster for: 6.8402 79.9998
📍 Nearby incidents found: 0
❌ No cluster found. Creating new cluster (CLS-006)
NEW INCIDENT: {
  user_id: new ObjectId('69f03beca1fb4905833b900e'),
  type: 'Flood',
  description: "<script>alert('xss')</script> Water rising rapidly near main street",
  image: null,
  location: { type: 'Point', coordinates: [ 79.9998, 6.8402 ] },
  assignedAuthorities: [],
  severity: 'Low',
  status: 'Pending',
  assigned_at: null,
  cluster_id: new ObjectId('6a77f50ceaff9af045775648'),
  verified_by: [],
  reported_inaccurate_by: [],
  _id: new ObjectId('6a77f50ceaff9af045775649'),
  status_history: [],
  timestamp: 2026-08-09T03:33:32.916Z
}
```

```
SAVED INCIDENT: {
  user_id: new ObjectId('69f03beca1fb4905833b900e'),
  type: 'Flood',
  description: "<script>alert('xss')</script> Water rising rapidly near main street",
  image: null,
  location: { type: 'Point', coordinates: [ 79.9998, 6.8402 ] },
  assignedAuthorities: [],
  severity: 'Low',
  status: 'Pending',
  assigned_at: null,
  cluster_id: new ObjectId('6a77f50ceaff9af045775648'),
  verified_by: [],
  reported_inaccurate_by: [],
  _id: new ObjectId('6a77f50ceaff9af045775649'),
  status_history: [],
  timestamp: 2026-08-09T03:33:32.916Z,
  __v: 0
}
👤 Auto-assigning responder "Fire Officer Nimal" (69c8f887ac2b0d97336825ce) to incident "6a77f50ceaff9af045775649"
✅ Assignment notifications created
✅ Status change notification created
POST /api/incidents 201 612 - 569.772 ms
[]
```

Field	Output Details
Test ID	SEC-007
Module	Security & Input Validation
Feature	Geospatial Boundary Validation (POST /api/incidents)
Actual Result	Out-of-bounds coordinates (lat: 195.0, lng: -200.0) bypassed express body parsing but were intercepted and rejected by the MongoDB geospatial engine (code 2: BadValue). The system safely aborted insertion and returned error details in 170.48 ms .
Status	PASS

POST http://localhost:5000/api/incidents

POST http://localhost:5000/api/incidents

POST http://localhost:5000/api/incidents

+

...

http://localhost:5000/api/incidents

Save

POST

http://localhost:5000/api/incidents

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

Cookies

Beautify

none

form-data

x-www-form-urlencoded

raw

binary

JSON

1

2

3

4

5

6

"type": "Accident",

"description": "Invalid GPS test submission",

"latitude": 195.0,

"longitude": -200.0

Body

Cookies

Headers (8)

Test Results

Status: 500 Internal Server Error

Time: 176 ms

Size: 491 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

1

2

3

4

"message": "Error creating incident",

"error": "invalid point in geo near query \$geometry argument: { type: \"Point\", coordinates: [-200, 195] } Longitude/latitude is out of bounds, lng: -200 lat: 195"


```

-----
Incoming API Request: [POST] /api/incidents
Time: 9:09:19 AM
-----
BODY: {
  type: 'Accident',
  description: 'Invalid GPS test submission',
  latitude: 195,
  longitude: -200
}
SEVERITY: undefined
Searching cluster for: 195 -200
ERROR: MongoServerError: invalid point in geo near query $geometry argument: { type: "Point", coordinates: [ -200, 195 ] } Longitude/latitude is out of bounds, lng: -200 lat: 195
    at Connection.sendCommand (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\mongodb\lib\cmap\connection.js:320:27)
    at process.processTicksAndRejections (node:internal/process/task_queues:105:5)
    at async Connection.command (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\mongodb\lib\cmap\connection.js:344:26)
    at async Server.command (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\mongodb\lib\sdam\server.js:208:29)
    at async tryOperation (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\mongodb\lib\operations\execute_operation.js:216:32)
    at async executeOperation (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\mongodb\lib\operations\execute_operation.js:80:16)
    at async FindCursor._initialize (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\mongodb\lib\cursor\find_cursor.js:62:26)
    at async FindCursor.cursorInit (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\mongodb\lib\cursor\abstract_cursor.js:620:27)
    at async FindCursor.fetchBatch (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\mongodb\lib\cursor\abstract_cursor.js:654:13)
    at async FindCursor.next (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\mongodb\lib\cursor\abstract_cursor.js:326:17) {
  errorLabelSet: Set(0) {},
  errorResponse: {
    ok: 0,
    errmsg: 'invalid point in geo near query $geometry argument: { type: "Point", coordinates: [ -200, 195 ] } Longitude/latitude is out of bounds, lng: -200 lat: 195',
    code: 2,
    codeName: 'BadValue',
  }
}

```

```

'$clusterTime': {
  clusterTime: new Timestamp({ t: 1786246758, i: 3 }),
  signature: [Object]
},
operationTime: new Timestamp({ t: 1786246758, i: 3 })
},
ok: 0,
code: 2,
codeName: 'BadValue',
'$clusterTime': {
  clusterTime: new Timestamp({ t: 1786246758, i: 3 }),
  signature: {
    hash: Binary.createFromBase64('/50/mI2EnmDvQxAJNyL77e/DiIM=', 0),
    keyId: new Long('7616398134586900483')
  }
},
operationTime: new Timestamp({ t: 1786246758, i: 3 })
}
POST /api/incidents 500 204 - 170.482 ms

```

Field	Output Details
Test ID	SEC-009
Module	Security & Input Validation
Feature	Oversized File Upload Restriction (POST /api/incidents)
Actual Result	File payload exceeding maximum size limits was intercepted by the multer middleware stack, throwing MulterError: File too large. The server safely aborted execution in 246.38 ms without consuming cloud storage or memory buffers.
Status	PASS

POST http://localhost:5000/api/

POST http://localhost:5000/api/

POST http://localhost:5000/api/

+

...

http://localhost:5000/api/incidents

Save

POST

http://localhost:5000/api/incidents

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

Key	Value	...	Bulk Edit
☐			
☒ image	Head_First_Agile.pdf		
Key	Value		

Body

Cookies

Headers (9)

Test Results

Status: 500 Internal Server Error

Time: 253 ms

Size: 1.71 KB

Save Response

Pretty

Raw

Preview

Visualize

HTML

```
4 <head>
5   <meta charset="utf-8">
6   <title>Error</title>
7 </head>
8
9 <body>
10  <pre>MulterError: File too large<br> &nbsp; &nbsp; &nbsp; at abortWithCode
    (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\multer\lib\make-middleware.js:101:22)
    at FileStream.<anonymous> (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\multer\lib\make-middleware.js:211:11)
    at FileStream.emit (node:events:508:20)<br> &nbsp; &nbsp; at SBMH.sscb [as _cb]
    (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\busboy\lib\types\multipart.js:479:32)
    at feed
    (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\streamsearch\lib\sbmh.js:248:10)<br>
    &nbsp; &nbsp; at SBMH.push
    (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\streamsearch\lib\sbmh.js:104:16)<br>
```

Incoming API Request: [POST] /api/incidents

Time: 9:18:50 AM

POST /api/incidents 500 1435 - 246.381 ms

MulterError: File too large

at abortWithCode (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\multer\lib\make-middleware.js:101:22)

at FileStream.<anonymous> (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\multer\lib\make-middleware.js:211:11)

at FileStream.emit (node:events:508:20)

at SBMH.sscb [as _cb] (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\busboy\lib\types\multipart.js:479:32)

at feed (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\streamsearch\lib\sbmh.js:248:10)

at SBMH.push (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\streamsearch\lib\sbmh.js:104:16)

at Multipart.write (G:\Projects\ResQNow-Crowdsourced_Location_Aware_Emergency_Response_System\backend\node_modules\busboy\lib\types\multipart.js:567:19)

at writeOrBuffer (node:internal\streams\writable:570:12)

at write (node:internal\streams\writable:499:10)

at Writable.write (node:internal\streams\writable:508:10)

Field	Output Details
Test ID	SEC-010
Module	Security & Input Validation
Feature	Document & Media File Upload Processing (POST /api/incidents)
Actual Result	Multipart form file attachment was received, uploaded securely to Cloudinary asset storage (resqnow_reports), and linked to new incident document 6a77fa54eaff9af045775655. Server returned HTTP 201 Created.
Status	PASS

POST http://localhost:5000/api/incidents

Save

POST

http://localhost:5000/api/incidents

Send

Params

Authorization

Headers (9)

Body

Pre-request Script

Tests

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

key	value
☒ type	Accident
☒ image	Analytics Module Test Cases.pdf
☒ description	Invalid image MIME type test
☒ latitude	6.8402
☒ longitude	79.9998
Key	Value

Body

Cookies

Headers (8)

Test Results

Status: 201 Created Time: 4.75 s Size: 950 B Save Response

Pretty

Raw

Preview

Visualize

JSON

```
17  "status": "Assigned",
18  "assigned_at": null,
19  "cluster_id": "6a77f50ceaff9af045775648",
20  "verified_by": [],
21  "reported_inaccurate_by": [],
22  "_id": "6a77fa54eaff9af045775655",
23  "status_history": [
24    {
25      "status": "Assigned",
26      "timestamp": "2024-05-09T09:56:01.587Z"
```

Field	Output Details
Test ID	SEC-011
Module	Security & Input Validation
Feature	Email Address Format Validation (POST /api/auth/register)
Input Data	{"name": "Test User", "email": "invalid-email format", "password": "Password123!", "role": "Citizen"}
Expected Result	API rejects registration attempt with HTTP 400 Bad Request and returns an explicit email format validation error.
Actual Result	Payload intercepted by express-validator middleware. Request blocked with HTTP 400 Bad Request ("Please provide a valid email address") in 43.67 ms .
Status	PASS

POST http://localhost:5000/api/auth/register

Save

POST

http://localhost:5000/api/auth/register

Send

Params

Authorization

Headers (8)

Body

Pre-request Script

Tests

Settings

none

form-data

x-www-form-urlencoded

raw

binary

JSON

Beautify

```
1  {
2    "name": "Test User",
3    "email": "invalid-email format",
4    "password": "Password123!",
5    "role": "Citizen",
6    "contact_number": "0775566931"
7  }
```

Body

Cookies

Headers (8)

Test Results

Status: 400 Bad Request

Time: 109 ms

Size: 445 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

```
3  "errors": [
4    {
5      "type": "field",
6      "value": "invalid-email format",
7      "msg": "Please provide a valid email address",
8      "path": "email",
9      "location": "body"
10   }
11 ]
12 }
```

```
-----
Incoming API Request: [POST] /api/auth/register
Time: 9:43:48 AM
-----
POST /api/auth/register 400 168 - 43.666 ms
```

Field	Output Details
Test ID	SEC-012
Module	Security & Input Validation
Feature	Password Length & Complexity Validation (POST /api/auth/register)
Input Data	{ "name": "Test User", "email": "validuser1@example.com", "password": "123", "role": "Citizen" }
Expected Result	API rejects registration attempt with HTTP 400 Bad Request due to insufficient password length (< 6 characters).
Actual Result	Payload intercepted by express-validator middleware prior to hashing or user persistence. Request blocked with HTTP 400 Bad Request ("Password must be at least 6 characters long") in 3.54 ms .
Status	PASS

POST http://localhost:5000/api/auth/register

Save

POST

http://localhost:5000/api/auth/register

Send

Params

Authorization

Headers (8)

Body

Pre-request Script

Tests

Settings

none

form-data

x-www-form-urlencoded

raw

binary

JSON

1

2

3

4

5

6

7

"name": "Test User",

"email": "validuser1@example.com",

"password": "123",

"role": "Citizen",

"contact_number": "077566931"

Body

Cookies

Headers (8)

Test Results

Status: 400 Bad Request

Time: 8 ms

Size: 438 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

1

2

3

4

5

6

7

8

9

10

"message": "Validation failed",

"errors": [

{

"type": "field",

"value": "123",

"msg": "Password must be at least 6 characters long",

"path": "password",

"location": "body"

}

]

Incoming API Request: [POST] /api/auth/register

Time: 9:46:41 AM

POST /api/auth/register 400 161 - 3.540 ms